

Computational Reproducibility Guide

Approach-Region Commitment and Environmental Friction:
A Cross-Phase Damping Atlas for Thermally Activated Chemical Reactions

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Overview

This guide documents every Python script required to reproduce the calculations, figures, and verification tests in the Atlas manuscript. All scripts are deposited at github.com/SymCUniverse/Chemistry and require only `numpy` and `matplotlib` (both pre-installed in Google Colab). No proprietary software is required.

Epistemic note. All χ_0 values are computed from independently measured or computed literature quantities via three estimator routes. No free parameters are used to place systems near $\chi_0 = 1$. The evidence tier (1–3) reflects how directly Γ_{eff} and Ω_0 are measured in the cited literature. The script `atlas_v1_provenance.py` documents every input source and demonstrates that regime assignments are not sensitive to small parameter variations.

Execution note. All scripts can be run in Google Colab with:

```
!python <script_name>.py
```

Figure scripts save `.svg` and `.png` automatically.

Purpose and Scope

Purpose. The ARC Atlas provides a phase-agnostic dynamical coordinate (χ_0) and a reproducible computational pathway for placing any thermally activated reaction on a shared friction axis using only independently measurable observables.

What χ_0 is not:

- χ_0 is **not** a rate predictor — it organizes the dynamical pre-exponential regime, not the full rate constant.
- χ_0 is **not** a universal law — it identifies a structural invariance in the linearized generator shared across phases.
- χ_0 is **not** fitted to reaction outcomes — all inputs are from independent measurements or computations.

How to compute χ_0 for a new system:

1. Identify the dominant approach-region mode frequency Ω_0 .
2. Measure or compute the dominant friction channel Γ_{eff} .

3. Choose the appropriate estimator route: Route A (spectroscopic linewidth), Route B (solvent τ_L), or Route C (electronic friction width).
4. Compute $\chi_0 = \Gamma_{\text{eff}}/(2\Omega_0)$ using the shared functions in `atlas_v1_constants.py`.
5. Place the system on the shared log-scale χ_0 axis and read off its dynamical regime from $\eta(\chi_0)$.

All numerical inputs are hardcoded in the scripts with inline source comments referencing the original literature. No external data download is required.

1 Canonical Constants and Shared Functions

`atlas_v1_constants.py`

Purpose: Single-source module for all physical constants and the three estimator-route functions. Every computation script derives from these definitions. Running the script standalone executes a 15-test self-verification suite.

Contents:

- Physical constants: $HC = hc = 1.23984193 \times 10^{-4}$ eV·cm; $c = 2.99792458 \times 10^{10}$ cm s $^{-1}$; conversion factors
- `chi_a(dnu_L, nu)` — Route A: $\chi_0 = \Delta\tilde{\nu}_L/(2\tilde{\nu})$
- `chi_b(tau_ps, lam_ev, dq_ang, m_amu, omega_s_cm1)` — Route B
- `chi_c(gamma_ev, nu_cm1)` — Route C: $\chi_0 = \Gamma_{\text{NN}}/(2hc\tilde{\nu})$
- `ivr_cm1(tau_s)` — IVR lifetime to linewidth: $\Gamma = 1/(2\pi c \tau_{\text{IVR}})$
- `eta(chi)` — $\eta(\chi) = 2\chi/(1 + \chi^2)$
- `regime(chi)` — classifies into underdamped / near-critical / overdamped

Scaling law verified: Route B scales as $\chi_0 \propto \Omega_s^2 \cdot \sqrt{m_{\text{eff}}}$ (15 PASS, 0 FAIL on self-test).

Core implementation (formula-critical lines):

```
# Route A: spectroscopic linewidth ratio
def chi_a(dnu_L, nu):          # Route A: spectroscopic ratio
    return dnu_L / (2 * nu)

# Route B: solvent friction (Zusman/GLE Markovian limit)
def chi_b(tau_ps, lam_ev, dq_ang, m_amu, ws_cm1):
    w0 = sqrt(2*lam_ev*EV_J / (m_amu*AMU_KG*(dq_ang*ANG_M)**2))
    ws = ws_cm1 * 2*pi*C_CMS          # angular frequency
    return (ws**2 * tau_ps*PS_S) / (2*w0)

# Route C: Newns-Anderson electronic friction
def chi_c(gamma_ev, nu_cm1):    # Route C: Newns-Anderson width
    return gamma_ev / (2 * HC * nu_cm1)

python atlas_v1_constants.py
```

2 Verification Scripts

`atlas_v1_verify_final.py`

Purpose: Verifies every equation and every χ_0 value in the manuscript that does not require physical experiment. **Must run before any submission; all 64 tests must pass.**

Test blocks:

- Block 1: Core functions — $\eta(\chi)$, window boundaries, small- χ approximation, hc constant, IVR lifetime conversion
- Block 2: Solid-state systems
- Block 3: Liquid-phase systems including HPTS D/H isotope analysis
- Block 4: Gas-phase systems
- Block 5: RRKM connection — P_{commit} , $N_{\text{encounters}}$, gas/liquid ratio
- Block 6: PCET isotope derivation
- Block 7: Nine-system table consistency

`python atlas_v1_verify_final.py`

Expected output: STATUS: ALL TESTS PASSED (64 PASS, 0 FAIL)

`atlas_v1_window_verify.py`

Purpose: Verifies the near-critical window and $\eta(\chi)$ function from mathematical first principles, independent of any literature values.

Covers:

- Analytic roots of $\eta(\chi) = 0.95$: schematic window [0.7239, 1.3813]; operational window [0.8, 1.3] (conservative)
- $P_{\text{commit}} = \eta(\chi_0)$ at all nine atlas systems
- Small- χ approximation accuracy: $|P - 2\chi|/P < 0.01\%$ for $\chi < 0.01$
- Physical constant cross-check: HC verified from h , c , e
- K-Fe exceptional-point value verified: $\chi_0 = 1.2825$

`python atlas_v1_window_verify.py`

`atlas_v1_gle_verify.py`

Purpose: Verifies the GLE derivation $\Gamma_{\text{eff}} = \Omega_s^2 \tau_L$ numerically by integrating the Ornstein-Uhlenbeck memory kernel $K(t) = \Omega_s^2 \exp(-t/\tau_L)$. Integration to $5\tau_L$ captures $> 99.3\%$ of the kernel area; the script integrates to $12\tau_L$ for the $< 10^{-4}$ relative-error tolerance check. Confirms that χ_0 computed from the integral matches the Route B formula to the same tolerance.

Core implementation:

```

# Memory kernel  $K(t) = \Omega_s^2 * \exp(-t / \tau_L)$ 
K_arr = omega_s**2 * np.exp(-t_arr / tau_L)

# Markovian limit:  $\text{integral} = \Omega_s^2 * \tau_L$ 
Geff_numerical = np.trapezoid(K_arr, t_arr)
Geff_analytic = omega_s**2 * tau_L
rel_err = abs(Geff_numerical - Geff_analytic) / Geff_analytic
# 5*tau captures >99.3% of kernel; script uses 12*tau for 1e-4 tolerance
assert rel_err < 1e-4 # confirms GLE derivation (Supplement S2)

python atlas_v1_gle_verify.py

```

atlas_v1_markovian_check.py

Purpose: Computes τ_L/τ_{rxn} for all Route B liquid-phase solvents and generates Supplement Table S1. Confirms that all Route B solvents satisfy $\tau_L \ll \tau_{\text{rxn}}$ (Markovian condition holds for the full solvent series). Slow solvents (EtOH, MeOH) are designated non-Debye upper bounds because $\tau_L > 1$ ps makes the single-Debye relaxation model approximate, not because the Markovian condition fails. Also verifies that relative solvent ordering is Tier 1 (bracket-independent).

Core implementation:

```

# All solvents satisfy  $\tau_L \ll \tau_{\text{rxn}}$  (Markovian condition)
# Slow-solvent designation reflects non-Debye relaxation,
# not Markovian failure -- see Supplement Table S1
DEBYE_RELIABILITY_PS = 1.0

for solvent, tau_ps in solvents:
    if tau_ps < DEBYE_RELIABILITY_PS:
        status = 'single-Debye estimate reliable'
    else:
        status = 'non-Debye slow-solvent -- upper bound'

# Key Tier-1 result: relative ordering is bracket-independent
ratio_numerical = chi_b(tau_A, ...) / chi_b(tau_B, ...)
ratio_analytical = tau_A / tau_B # all other params cancel
assert abs(ratio_numerical - ratio_analytical) < 1e-6

python atlas_v1_markovian_check.py

```

atlas_v1_deltaQ_sensitivity.py

Purpose: Verifies that $\chi_0 \propto \Delta Q$ (linear scaling, not quadratic) and generates Supplement Table S3 showing χ_0 for $\text{Fe}(\text{CN})_6/\text{MeCN}$ at $\Delta Q = 0.28, 0.40, 0.52$ Å at central m_{eff} and Ω_s . Also documents the full three-parameter bracket, including the all-minimum corner ($\chi_0 = 0.15$, underdamped), explaining the Tier 2 absolute designation.

Core implementation:

```

# chi_0 = Omega_s^2 * tau_L / (2 * Omega_0)
# Omega_0 = sqrt(2*lambda / (m * DQ^2)) => Omega_0 ~ 1/DQ
# Therefore chi_0 ~ DQ (linear, not quadratic)

chi_28 = chi_b(tau, lam, 0.28, m, ws) # -30% DQ
chi_40 = chi_b(tau, lam, 0.40, m, ws) # central
chi_52 = chi_b(tau, lam, 0.52, m, ws) # +30% DQ

# Verify linearity
assert abs(chi_28/chi_40 - 0.28/0.40) < 1e-5 # PASS
assert abs(chi_52/chi_40 - 0.52/0.40) < 1e-5 # PASS

python atlas_v1_deltaQ_sensitivity.py

```

3 Computation Scripts by Phase

atlas_v1_solids.py

Purpose: Computes χ_0 for the three solid-state systems.

Inputs and sources:

1. $\text{N}_2/\text{K-Fe}(111)$ — Route C, Tier 2. $\Gamma_{\text{NN}} = 0.45$ eV, $\tilde{\nu} = 1415$ cm^{-1} . Source: DFT d-band calculations from the companion paper (Christensen 2026 ChemRxiv, DOI: [10.26434/chemrxiv.15004963/v1](https://doi.org/10.26434/chemrxiv.15004963/v1)); methodology from Maurer et al. PRB **94**, 115432 (2016).
2. $\text{CO}/\text{Cu}(111)$ — Route A, Tier 1. $\Delta\tilde{\nu}_L = 10\text{--}18$ cm^{-1} , $\tilde{\nu}_\perp = 344$ cm^{-1} . Source: Hirschmugl et al. PRL **65**, 480 (1990); Persson & Ryberg PRB **32**, 3586 (1985).
3. Li^+/LLZO — Route C-analog, Tier 3. $\Gamma_{\text{ph}} = 40\text{--}80$ cm^{-1} , $\tilde{\nu}_0 = 234$ cm^{-1} . Source: Kozinsky et al. PRL **116**, 055901 (2016) (MD); Wood et al. PRMater **2**, 065401 (2018) (neutron).

python atlas_v1_solids.py

atlas_v1_liquids.py

Purpose: Computes χ_0 for the three liquid-phase systems. Includes solvent series tables for ET and SN1 (relative ordering Tier 1) and HPTS D/H ratio.

Inputs and sources:

1. $\text{Fe}(\text{CN})_6^{3-/4-}$ ET — Route B, Tier 1r/2a. $\lambda = 0.68$ eV, $\Delta Q = 0.40$ Å, $m_{\text{eff}} = 150$ amu (central), $\Omega_s = 100$ cm^{-1} . τ_L series: Jimenez et al. Nature **369**, 471 (1994). λ : Grampp & Jaenicke Ber. Bunsenges. **95**, 904 (1991).
2. $t\text{-BuCl}$ SN1 — Route B, Tier 1r/2a. $\lambda = 0.52$ eV, $\Delta Q = 0.30$ Å, $m_{\text{eff}} = 120$ amu. Rate order: Winstein & Fainberg JACS **78**, 2770 (1956).
3. HPTS PT — Routes A and B, Tier 2/diag. $\tilde{\nu}_{\text{OH}^*} = 2750$ cm^{-1} , $\Delta\tilde{\nu}_L = 160$ cm^{-1} . $m_H = 1.008$ amu (proton), $m_D = 2.014$ amu (deuteron). Source: Spry & Fayer JCP **127**, 204501 (2007). Route A is diagnostic only (Lorentzian fraction $\approx 40\% < 50\%$ threshold).

python atlas_v1_liquids.py

atlas_v1_gases.py

Purpose: Computes χ_0 for the three gas-phase systems. All gas-phase systems use IVR lifetime broadening or predissociation linewidths as the effective friction channel (structural analogy; see the manuscript section “Results III: Gas-Phase Systems”).

IVR conversion:

$$\Gamma_{\text{IVR}} \text{ (cm}^{-1}\text{)} = \frac{1}{2\pi c \tau_{\text{IVR}}}, \quad c = 2.99792458 \times 10^{10} \text{ cm s}^{-1} \quad (1)$$

Systems:

1. Cyclopropane $\rightarrow$ propene — Route A (IVR), Tier 2. $\tilde{\nu}_{\text{ring}} = 1070 \text{ cm}^{-1}$, $\Gamma_{\text{IVR}} = 3\text{--}15 \text{ cm}^{-1}$. Source: Rabinovitch & Setser Adv. Photochem. **3**, 1 (1964).
2. $\text{NO}_2 \rightarrow \text{NO} + \text{O}$ — Route A (Lorentzian), Tier 1. $\tilde{\nu}_{\text{app}} = 750\text{--}850 \text{ cm}^{-1}$, $\Gamma_{\text{pd}} = 3\text{--}20 \text{ cm}^{-1}$ (directly measured). Source: Ionov et al. JCP **97**, 9486 (1992).
3. $\text{CH}_3\text{NC} \rightarrow \text{CH}_3\text{CN}$ — Route A (IVR), Tier 2. $\tau_{\text{IVR}} = 3 \text{ ps} \rightarrow \Gamma = 1.770 \text{ cm}^{-1}$, $\tilde{\nu}_{\text{app}} = 460 \text{ cm}^{-1}$. Source: Reddy & Berry Chem. Phys. Lett. **66**, 223 (1979).

python atlas_v1_gases.py

atlas_v1_rrkm.py

Purpose: Verifies the gas-phase RRKM connection. Computes $P_{\text{commit}} = \eta(\chi_0)$ and $N_{\text{encounters}} \approx 1/(2\chi_0)$ for all three gas-phase systems, and confirms the gas/liquid χ_0 ratio spans ≈ 2.6 decades.

Key outputs:

System	χ_0	P_{commit}	N_{enc}
CH_3NC	0.0019	0.0038	≈ 260
Cyclopropane	0.0037	0.0075	≈ 134
NO_2	0.0063	0.0126	≈ 80

Note: N_{enc} values are classical structural-analogy estimates, not quantum-mechanical predictions.

python atlas_v1_rrkm.py

4 Extended Analysis Scripts

atlas_v1_route_b_full.py

Purpose: Complete Route B analysis including:

1. All six solvents for both $\text{Fe}(\text{CN})_6$ ET and $t\text{-BuCl}$ SN1, including heavily overdamped EtOH ($\chi_0 \approx 93$ for SN1)
2. Full $m_{\text{eff}} \times \Omega_s$ bracket grid for $\text{Fe}(\text{CN})_6/\text{MeCN}$
3. Algebraic proof that the HPTS D/H ratio is independent of the absolute Route B bracket
4. Verification of the Route B scaling law: $\chi_0 \propto \Omega_s^2 \cdot \sqrt{m_{\text{eff}}}$

Bracket independence proof (key result):

$$\frac{\chi_0^D}{\chi_0^H} = \frac{\tau_L(\text{D}_2\text{O})}{\tau_L(\text{H}_2\text{O})} \cdot \sqrt{\frac{m_D}{m_H}} = 1.5714 \times 1.4135 = 2.2212 \quad (2)$$

The collective-shell parameters ($m_{\text{eff}}^{\text{shell}}$, ΔQ , λ , Ω_s) are identical for H and D and cancel in the ratio. This is verified numerically across a 4×4 bracket grid.

Core implementation:

```
# Proton and deuteron masses (not scaled m_eff)
m_H, m_D = 1.008, 2.014    # amu
ratio_predicted = (tau_D20 / tau_H20) * sqrt(m_D / m_H)    # = 2.2212

# Ratio is independent of lam, dq, ws -- verify across full bracket
for lam_i in [0.58, 0.68, 0.78]:    # eV bracket
    for dq_i in [0.28, 0.40, 0.52]:    # Angstrom bracket
        for ws_i in [50, 100, 150]:    # cm-1 bracket
            chi_H = chi_b(tau_H20, lam_i, dq_i, m_H, ws_i)
            chi_D = chi_b(tau_D20, lam_i, dq_i, m_D, ws_i)
            ratio = chi_D / chi_H
            assert abs(ratio - ratio_predicted) < 1e-6    # PASS all 27

python atlas_v1_route_b_full.py
```

atlas_v1_provenance.py

Purpose: Input provenance and anti-forcing demonstration. For every system, lists the exact literature source for each input parameter, computes χ_0 , and asks how much the primary input must change to cross the nearest regime boundary.

Anti-forcing evidence in the output:

- Only $\text{N}_2/\text{K-Fe(111)}$ is inside the near-critical window $[0.8, 1.3]$. All other systems are underdamped or overdamped.
- Gas-phase systems are underdamped by 2–3 orders of magnitude. No physically plausible Γ_{IVR} adjustment moves them into the near-critical window.
- Liquid-phase systems (MeCN , H_2O) are slightly overdamped ($\chi_0 = 1.44\text{--}1.63$), not inside the window.
- $\text{CO}/\text{Cu(111)}$ and LLZO are both underdamped. A forced dataset would not include these.

python atlas_v1_provenance.py

5 Figure Scripts

All figure scripts save **.svg** (vector, Illustrator-compatible) and **.png** (300 dpi) automatically on completion. No input files are required; all data are hardcoded with inline comments referencing the verification tests. Manuscript **.pdf** figures are exported from the generated **.svg** files after final label adjustment in Adobe Illustrator; the **.svg** outputs are the primary reproducible artifacts.

Main figures

- `atlas_fig1_crossphase.py` — Cross-phase damping atlas (Fig. 1). All nine systems on one log-scale χ_0 axis. Marker shape = evidence grade. Marker color = phase. Error bars show full uncertainty brackets.
- `atlas_fig2_eta_curve.py` — Dynamical efficiency landscape (Fig. 2). $\eta(\chi_0) = 2\chi_0/(1 + \chi_0^2)$ on log scale with all nine systems overlaid. Y-axis label reads “Schematic dynamical efficiency projection” to make clear this is not normalized experimental performance.
- `atlas_fig3_routes.py` — Estimator route data (Fig. 3), two panels: (a) Route B: full solvent series for $\text{Fe}(\text{CN})_6$ ET and *t*-BuCl SN1 (six solvents each, log-scale bars). (b) Route C: K-Fe surface promotion series ($\text{Fe}(111) \rightarrow \text{Fe}(110) \rightarrow \text{K-Fe}(111)$).
- `atlas_fig4_ladder.py` — Cross-phase environmental friction ladder (Fig. 4). Four phase lanes (Gas, Liquid, Surface, Solid) on shared log χ_0 axis with friction mechanism annotations. Synthesis figure; makes the five-order-of-magnitude progression visible.

Supplementary figures

- `atlas_figS1_sensitivity.py` — Route B sensitivity heatmap (Fig. S1). χ_0 vs. $m_{\text{eff}} \times \Omega_s$ grid for $\text{Fe}(\text{CN})_6/\text{MeCN}$. Near-critical contours at $\chi_0 = 0.8$ and 1.3 overlaid. Demonstrates that absolute placement is Tier 2 while relative ordering is Tier 1.
- `atlas_figS2_hpts.py` — HPTS bifurcated perturbation (Fig. S2). Bar chart showing proton-mass factor (1.414), solvent τ_L isotope factor (1.571), combined D/H ratio (2.221), and approach-region upper bound ($\sqrt{2} = 1.414$).
- `atlas_figS3_gasphase.py` — Gas-phase low- χ_0 limit (Fig. S3). Log-log plot of P_{commit} vs. χ_0 with all three gas-phase systems and N_{enc} annotations.
- `atlas_v1_ivr_statistical_limit.py` — Bixon-Jortner quantum-to-classical correspondence (Fig. S4). Three-panel figure showing quantum survival probability vs. classical exponential envelope for each gas-phase system, with parameters calibrated to the atlas Γ_{IVR} values. Tier 1 (NO_2): exact continuum-coupling convergence. Tier 2 (cyclopropane, CH_3NC): sparse state density with visible quantum recurrences; RRKM thermal ensemble recovers the smooth $\exp(-\Gamma t)$ envelope. Parameters: $V = 1.0 \text{ cm}^{-1}$ with ρ set by Fermi’s Golden Rule.

Core implementation (Hamiltonian + time evolution):

```
# Fermi's Golden Rule: rho = Gamma_IVR / (2*pi*V^2)
rho = Gamma_cm1 / (2 * pi * V_cm1**2)

# Dark-state energies on a deterministic uniform grid (seed-free)
# => Fig. S4 is exactly reproducible without random number generation
H[0, 0] = 0.0          # bright state energy
H[i, i] = dark_energies[i-1] # dark state energies (uniform grid)
H[0, i] = H[i, 0] = V   # uniform coupling

# Quantum time evolution: P(t) = |<0|psi(t)>|^2
```



```

eigvals, eigvecs = np.linalg.eigh(H) # real symmetric => real eigvecs
weights = np.abs(eigvecs[0, :])**2 # |<0|k>|^2, safe for general use
amplitude = np.sum(weights * np.exp(-1j * omega_k * t))
P_quantum = np.abs(amplitude)**2

# Classical envelope for comparison
P_classical = exp(-Gamma_cm1 * 2*pi*c_cm_ps * t)

```

6 Evidence Grade Summary

System	Phase	Route	χ_0	Tier	Regime
N ₂ /K-Fe(111)	Solid	C	1.2825	2	near-critical
CO/Cu(111)	Solid	A	0.019	1	underdamped
Li ⁺ /LLZO	Solid	C-ph	0.12	3	underdamped
Fe(CN) ₆ /MeCN	Liquid	B	1.52	1r/2a	overdamped
<i>t</i> -BuCl/H ₂ O	Liquid	B	1.63	1r/2a	overdamped
HPTS/H ₂ O	Liquid	A+B	1.44(B)	2/diag	overdamped
Cyclopropane→propene	Gas	A(IVR)	0.0037	2	underdamped
NO ₂ →NO+O	Gas	A(Lor)	0.0063	1	underdamped
CH ₃ NC→CH ₃ CN	Gas	A(IVR)	0.0019	2	underdamped

1r/2a = Tier 1 relative; Tier 2 absolute. 2/diag = Tier 2 absolute; Route A diagnostic (< 50% Lorentzian).

Tier definitions:

- Tier 1: both Γ_{eff} and Ω_0 directly measured
- Tier 2: one measured, one computed or bracketed from literature
- Tier 3: both quantities from simulation or prospective model; for bracketing only (e.g., LLZO MD friction kernel). RRKM-inferred gas-phase IVR widths are Tier 2 when paired with a measured approach frequency

7 Complete Reproduction Workflow

Run in this order for a clean end-to-end check. All scripts must complete without errors.

1. Canonical constants and functions (15-test self-check)

```
python atlas_v1_constants.py
```

2. Window and eta function verification (math only, no literature)

```
python atlas_v1_window_verify.py
```

3. Master algebraic verification suite (64 tests)

```
python atlas_v1_verify_final.py
```

4. Input provenance and anti-forcing demonstration

```
python atlas_v1_provenance.py
```

```

# 5. Solid-state chi values
python atlas_v1_solids.py

# 6. Liquid-phase chi values (solvent series)
python atlas_v1_liquids.py

# 7. Gas-phase chi values
python atlas_v1_gases.py

# 8. RRKM connection
python atlas_v1_rrkm.py

# 9. Complete Route B analysis (brackets + HPTS D/H proof)
python atlas_v1_route_b_full.py

# 10. GLE derivation verification
python atlas_v1_gle_verify.py

# 11. Markovian validity check (Supplement Table S1)
python atlas_v1_markovian_check.py

# 12. Delta Q sensitivity (Supplement Table S3)
python atlas_v1_deltaQ_sensitivity.py

# 13. Main figures (saves .svg and .png)
python atlas_fig1_crossphase.py
python atlas_fig2_eta_curve.py
python atlas_fig3_routes.py
python atlas_fig4_ladder.py

# 14. Supplementary figures (saves .svg and .png)
python atlas_figS1_sensitivity.py
python atlas_figS2_hpts.py
python atlas_figS3_gasphase.py
python atlas_v1_ivr_statistical_limit.py

```

Required clean exits: atlas_v1_verify_final.py: 64 PASS, 0 FAIL. atlas_v1_constants.py: 15 PASS, 0 FAIL. atlas_v1_gle_verify.py: all GLE PASS. atlas_v1_markovian_check.py: all checks PASS. atlas_v1_deltaQ_sensitivity.py: all checks PASS. All figure scripts: prints Saved: with both output filenames.

Data Availability

All scripts are deposited at github.com/SymCUniverse/Chemistry.

Companion paper preprint (Approach-Region Dynamics): doi.org/10.26434/chemrxiv.15004963/v1

StabilityArch (published): doi.org/10.1038/s41598-026-56887-7

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